

Pythagoras, mensuration and 3D - Set 1 - Solutions

Specification M5.7, M5.11-M5.13, M5.15-M5.17.

Answer key

Question	Answer	Topic
1	B	Pythagoras, mensuration and 3D
2	C	Pythagoras, mensuration and 3D
3	E	Pythagoras, mensuration and 3D
4	F	Pythagoras, mensuration and 3D
5	B	Pythagoras, mensuration and 3D
6	E	Pythagoras, mensuration and 3D
7	D	Pythagoras, mensuration and 3D
8	F	Pythagoras, mensuration and 3D
9	E	Pythagoras, mensuration and 3D
10	B	Pythagoras, mensuration and 3D
11	F	Pythagoras, mensuration and 3D
12	F	Pythagoras, mensuration and 3D

Worked solutions

Question 1 MEN-01 · Pythagoras, mensuration and 3D · spec M5.7

A cuboid measures 3 cm by 4 cm by 12 cm. Find the length of its longest diagonal.

Answer: B

$$\sqrt{9 + 16 + 144} = \sqrt{169} = 13.$$

Full method - how and why

The space diagonal of a cuboid: Pythagoras applied twice, which gives the three-dimensional formula.

- 1 First find the diagonal of the base: $\sqrt{3^2 + 4^2} = 5$.

why: That is a 3-4-5 triangle - the base diagonal lies in a horizontal plane.

- 2 Now use that with the height: $\sqrt{5^2 + 12^2} = \sqrt{169} = 13$.

why: The space diagonal, the base diagonal and the vertical edge form a second right-angled triangle. This is a 5-12-13 triple.

- 3 Equivalently, $\sqrt{a^2 + b^2 + c^2} = \sqrt{9 + 16 + 144} = 13$ cm.

why: The two applications of Pythagoras combine into one formula.

Where it goes wrong

- Using only two of the three dimensions, giving 5 or $\sqrt{153}$.
- Adding the lengths: $3 + 4 + 12 = 19$.

Takeaway Space diagonal = $\sqrt{a^2 + b^2 + c^2}$ - Pythagoras twice in one step.

Question 2 MEN-03 · Pythagoras, mensuration and 3D · spec M5.16

A trapezium has parallel sides of length 8 cm and 14 cm and area 77 cm^2 . Find the perpendicular distance between the parallel sides.

Answer: C

$$\frac{1}{2}(8 + 14)h = 11h = 77 \Rightarrow h = 7.$$

Full method - how and why

Trapezium area formula, rearranged for the height.

- 1 Write the formula: area = $\frac{1}{2}(a + b)h$ where a, b are the parallel sides.

why: The height is the **perpendicular** distance between the parallel sides.

- 2 Substitute: $\frac{1}{2}(8 + 14)h = 11h$.

why: $\frac{1}{2} \times 22 = 11$.

- 3 Solve: $11h = 77$, so $h = 7$ cm.

why: Divide both sides by 11.

Where it goes wrong

- Forgetting the $\frac{1}{2}$ and answering 3.5 - an offered option.
- Using the difference of the parallel sides instead of the sum.

Takeaway Trapezium: average the parallel sides, then multiply by the perpendicular height.

Question 3 MEN-04 · Pythagoras, mensuration and 3D · spec M5.16

A cone has base radius 6 cm and slant height 10 cm. Find its volume. (Volume of a cone = $\frac{1}{3}\pi r^2 h$.)

Answer: E

$$\text{Height} = \sqrt{100 - 36} = 8. \text{ Volume} = \frac{1}{3}\pi \cdot 36 \cdot 8 = 96\pi.$$

Full method - how and why

The cone's slant height, radius and vertical height form a right-angled triangle. Find the height, then use the volume formula.

- 1 Use Pythagoras: $h = \sqrt{l^2 - r^2} = \sqrt{100 - 36} = 8$ cm.
why: The slant height is the **hypotenuse**, so subtract; this is another 6-8-10 triangle.
- 2 Apply the given formula: $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(8)$.
why: The formula is provided in the question, as the specification promises.
- 3 Compute: $\frac{1}{3} \times 288 = 96$, so $V = 96\pi \text{ cm}^3$.
why: $36 \times 8 = 288$.

Where it goes wrong

- Using the slant height as the vertical height, giving 120π - an offered option.
- Forgetting the $\frac{1}{3}$ and answering 288π .

Takeaway Cone: $\text{slant}^2 = r^2 + h^2$; and the volume carries a factor of $\frac{1}{3}$ compared with a cylinder.

Question 4 MEN-05 · Pythagoras, mensuration and 3D · spec M5.17

A sphere and a cube have the same volume. The cube has side length a . Find the radius of the sphere. (Volume of a sphere = $\frac{4}{3}\pi r^3$.)

Answer: F

$$\frac{4}{3}\pi r^3 = a^3 \Rightarrow r^3 = \frac{3a^3}{4\pi} \Rightarrow r = a\sqrt[3]{\frac{3}{4\pi}}.$$

Full method - how and why

Equate the two volumes and solve for r - an algebraic rearrangement with a cube root.

- 1 Set the volumes equal: $\frac{4}{3}\pi r^3 = a^3$.
why: The sphere's formula is given; the cube's is a^3 .
- 2 Isolate r^3 : $r^3 = \frac{3a^3}{4\pi}$.
why: Multiply both sides by $\frac{3}{4\pi}$.
- 3 Take the cube root: $r = a\sqrt[3]{\frac{3}{4\pi}}$.
why: The cube root of a^3 is a , and the constant keeps its cube root.

Where it goes wrong

- Inverting the fraction, giving $a\sqrt[3]{\frac{4\pi}{3}}$ - an offered option.
- Taking a square root instead of a cube root.

Takeaway Equate volumes, isolate r^3 , then cube-root everything including the constant.

Question 5 MEN-06 · Pythagoras, mensuration and 3D · spec M5.16

A square of side 10 has a quarter-circle of radius 10 drawn inside it, centred at one corner. Find the area of the square not covered by the quarter-circle.

Answer: B

$$100 - \frac{1}{4}\pi \cdot 100 = 100 - 25\pi.$$

Full method - how and why

A composite area: subtract the quarter-circle from the square.

1 Square area: $10^2 = 100$.

why: Side squared.

2 Quarter-circle area: $\frac{1}{4}\pi r^2 = \frac{1}{4}\pi(100) = 25\pi$.

why: The radius equals the side, and a quarter of a circle is $\frac{1}{4}$ of πr^2 .

3 Subtract: $100 - 25\pi$.

why: Leave π exact - no calculator is available anyway.

Where it goes wrong

- Forgetting the quarter and subtracting 100π , which would give a negative area.
- Using $\frac{1}{4}\pi r$ (circumference) instead of area.

Takeaway Composite areas: subtract the parts, and check the answer is positive ($25\pi \approx 78.5 < 100$ ✓).

Question 6 MEN-07 · Pythagoras, mensuration and 3D · spec M5.7

A ladder of length 6.5 m leans against a vertical wall with its foot 2.5 m from the wall on horizontal ground. How far up the wall does it reach?

Answer: E

$$\sqrt{6.5^2 - 2.5^2} = \sqrt{42.25 - 6.25} = \sqrt{36} = 6.$$

Full method - how and why

A ladder against a wall is a right-angled triangle with the ladder as hypotenuse.

1 Identify the hypotenuse: the ladder, 6.5 m - the longest side, opposite the right angle at the base of the wall.

why: The wall and ground are perpendicular.

2 Apply Pythagoras: height = $\sqrt{6.5^2 - 2.5^2} = \sqrt{42.25 - 6.25}$.

why: Subtract, because we want a shorter side.

3 Compute: $\sqrt{36} = 6$ m.

why: Decimals chosen to give a whole number - a sign the working is right.

Where it goes wrong

- Adding the squares, giving $\sqrt{48.5}$ - an offered option.
- Arithmetic: $6.5^2 = 42.25$, not 36.25.

Takeaway Hypotenuse known $\Rightarrow$ subtract squares; hypotenuse unknown $\Rightarrow$ add them.

Question 7 MEN-08 · Pythagoras, mensuration and 3D · spec M5.18

In a right-angled triangle the hypotenuse is 10 and one angle is 30° . Find the area of the triangle.

Answer: D

$$\text{Sides } 10\sin 30^\circ = 5 \text{ and } 10\cos 30^\circ = 5\sqrt{3}. \text{ Area} = \frac{1}{2} \cdot 5 \cdot 5\sqrt{3} = \frac{25\sqrt{3}}{2}.$$

Full method - how and why

Right-angled trigonometry to get the two legs, then the standard triangle area.

- 1 The side opposite the 30° angle: $10\sin 30^\circ = 10 \times \frac{1}{2} = 5$.

why: $\sin = \frac{\text{opposite}}{\text{hypotenuse}}$, so opposite = hypotenuse $\times \sin$.

- 2 The adjacent side: $10\cos 30^\circ = 10 \times \frac{\sqrt{3}}{2} = 5\sqrt{3}$.

why: Exact values must be recalled: $\cos 30^\circ = \frac{\sqrt{3}}{2}$.

- 3 Area = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2}(5)(5\sqrt{3}) = \frac{25\sqrt{3}}{2}$.

why: The two legs are perpendicular, so they serve as base and height.

Where it goes wrong

- Forgetting the $\frac{1}{2}$, giving $25\sqrt{3}$ - an offered option.
- Using the hypotenuse as one of the perpendicular sides.

Takeaway In a right-angled triangle the two legs are base and height; get them from $\sin$ and $\cos$ of the hypotenuse.

Question 8 MEN-09 · Pythagoras, mensuration and 3D · spec M5.16

The volume of a cube is numerically equal to its surface area. Find the length of its side.

Answer: F

$$a^3 = 6a^2 \Rightarrow a = 6.$$

Full method - how and why

Set the two expressions equal and solve. The equation is cubic but factorises immediately.

- 1 Write both: volume = a^3 , surface area = $6a^2$ (six square faces).

why: A cube has six identical faces of area a^2 .

- 2 Set them equal: $a^3 = 6a^2$.

why: 'Numerically equal' means the numbers match, though the units differ.

- 3 Factorise rather than divide: $a^2(a - 6) = 0$, so $a = 0$ or $a = 6$. A cube must have $a > 0$, so $a = 6$.

why: Dividing by a^2 is fine here (a side length is non-zero), but factorising is the safer habit.

Where it goes wrong

- Taking a square root somewhere and answering $\sqrt{6}$ - an offered option.
- Using $4a^2$ or a^2 for the surface area.

Takeaway Cube: $V = a^3$, $S = 6a^2$; equate and factorise rather than dividing by a variable.

Question 9 MEN-11 · Pythagoras, mensuration and 3D · spec M5.16

Which of the following statements is/are true?

- I** Doubling the radius of a circle doubles its circumference.
- II** Doubling the radius of a sphere multiplies its volume by 8.
- III** Doubling the radius of a cylinder while halving its height leaves its volume unchanged.

Answer: E

$C = 2\pi r$ is linear in r (I). $V \propto r^3$ (II). $\pi(2r)^2 \cdot \frac{h}{2} = 2\pi r^2 h$: doubled, not unchanged (III false).

Full method - how and why

Three scaling claims. Each is checked by substituting the new dimensions into the formula.

- 1** I: $C = 2\pi r$ is linear in r , so doubling r doubles C . **True.**
why: Circumference scales with the first power of the radius.
- 2** II: $V = \frac{4}{3}\pi r^3$, so doubling r multiplies the volume by $2^3 = 8$. **True.**
why: Volume scales with the cube.
- 3** III: new volume $= \pi(2r)^2 \left(\frac{h}{2}\right) = \pi(4r^2) \left(\frac{h}{2}\right) = 2\pi r^2 h$ - **double**, not unchanged. **False.**
why: The radius is squared, so doubling it quadruples that factor; halving the height only removes a factor of 2.
- 4** I and II only.
why: Match to the option list.

Where it goes wrong

- Assuming the doubling and halving in III cancel - the radius is squared, so they do not.
- Confusing area and volume scaling in II.

Takeaway Substitute the new dimensions into the formula; squared quantities change faster than linear ones.

Question 10 MEN-12 · Pythagoras, mensuration and 3D · spec M5.16

A running track consists of two straights of length 100 m and two semicircular ends of radius r . The total length of the track is 400 m. Find r .

Answer: B

$$200 + 2\pi r = 400 \Rightarrow r = \frac{100}{\pi}.$$

Full method - how and why

The two semicircular ends together make one full circle. Write the perimeter and solve.

- 1** Total length = two straights + two semicircles $= 2(100) + 2(\pi r)$.
why: A semicircular arc has length πr ; two of them make a full circumference $2\pi r$.
- 2** Set up: $200 + 2\pi r = 400$.
why: Substitute the given total.
- 3** Solve: $2\pi r = 200$, so $r = \frac{100}{\pi}$.
why: Divide by 2π .

Where it goes wrong

- Using πr for both ends together, giving $\frac{200}{\pi}$ - an offered option.
- Using the area formula instead of arc length.

Takeaway Two semicircles of the same radius make one circle: total arc $= 2\pi r$.

Question 11 MEN-14 · Pythagoras, mensuration and 3D · spec M5.16

A cylindrical tin of radius 4 cm and height 10 cm is full of water. The water is poured into an empty cylindrical jug of radius 5 cm. How deep is the water in the jug?

Answer: F

$$\pi \cdot 16 \cdot 10 = \pi \cdot 25 \cdot h \Rightarrow h = \frac{160}{25} = 6.4.$$

Full method - how and why

Conservation of volume: the water keeps its volume, so equate the two cylinder volumes.

1 Volume in the tin: $\pi r^2 h = \pi(16)(10) = 160\pi$.

why: $r = 4$ so $r^2 = 16$.

2 In the jug the same volume occupies $\pi(25)h$.

why: Radius 5, unknown depth h ; the π will cancel.

3 Equate and solve: $160\pi = 25\pi h \Rightarrow h = \frac{160}{25} = 6.4$ cm.

why: The wider jug means a shallower depth - a good sanity check.

Where it goes wrong

- Using the radii rather than their squares, giving 8 cm - an offered option.
- Expecting the depth to increase because the jug is 'bigger'.

Takeaway Pouring conserves volume: equate $\pi r_1^2 h_1 = \pi r_2^2 h_2$; a wider container gives a shallower depth.

Question 12 MEN-15 · Pythagoras, mensuration and 3D · spec M5.17

A solid consists of a hemisphere of radius 3 on top of a cylinder of radius 3 and height 4. Find its volume. (Sphere volume $\frac{4}{3}\pi r^3$.)

Answer: F

$$\text{Cylinder } \pi \cdot 9 \cdot 4 = 36\pi; \text{ hemisphere } \frac{1}{2} \cdot \frac{4}{3}\pi \cdot 27 = 18\pi. \text{ Total } 54\pi.$$

Full method - how and why

A composite solid: add the two volumes, halving the sphere formula for the hemisphere.

1 Cylinder: $\pi r^2 h = \pi(9)(4) = 36\pi$.

why: $r = 3$, $h = 4$.

2 Hemisphere: half a sphere, $\frac{1}{2} \times \frac{4}{3}\pi r^3 = \frac{2}{3}\pi(27) = 18\pi$.

why: $\frac{2}{3} \times 27 = 18$; the sphere formula is given if needed.

3 Add: $36\pi + 18\pi = 54\pi$.

why: The solids sit on top of each other, so volumes simply add.

Where it goes wrong

- Using the full sphere, giving 72π - an offered option.
- Computing r^3 as 9 instead of 27.

Takeaway Composite solids: compute each piece separately and add; a hemisphere is exactly half a sphere.